

# Longitudinal media review for audition-oriented violin practice

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High-level music training produces dense evidence: daily practice videos, repeated repertoire, changing technical constraints, and periodic performance goals. Most of that evidence is difficult to use because it is stored as unstructured media rather than as a longitudinal record. We report Curtis Media Review, a live AO Labs system that indexes public practice media, identifies practice candidates, records model-assisted section reviews, and converts the current state into a compact practice record. The checked live state on 7 May 2026 used the YouTube channel @nalalan as the primary source, indexed 207 public videos, marked 102 as practice candidates, marked 82 as long-form candidates, stored 20 media samples, retained 51 reviewed sections, and recorded 80 Curtis-focused findings. The first screen reports the repertoire-identification state, the score state, one immediate tip, and a source-linked highlight clip so the label can be checked against the actual playing. A growing repertoire list stores day-level progress entries, while a dated practice log keeps clips, detection status, and sample counts visible without exposing the full raw record by default. Possible labels, user-rejected false labels, weak evidence, and readiness scores inferred only from piece identification must remain distinguishable from confirmed repertoire and judged playing quality. The system uses GPT-5.5 for vision and text review and separate audio models for identification and verification, while preserving explicit limits: Curtis admission cannot be predicted from current samples, weak audio/video evidence is marked unjudged, and the system reports practice signals rather than admissions probability. The contribution is an experimental method for turning accumulated practice media into auditable feedback loops, progress history, and source-bounded coaching suggestions.

## 1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (1, 2). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis Media Review addresses this problem as a live software system rather than as a portfolio page (3). The system indexes public practice media, classifies candidate practice logs, records technical observations, and keeps the current practice constraint visible. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and return a small number of practice constraints that can be tested in subsequent recordings.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (4). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

## 2 System state

The first production state used the live backend at <https://curtis.aolabs.io/api/curtis/media-status>. The checked state on 7 May 2026 reported the values in Table 1. These are operating facts, not outcome claims.

### 2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. Other videos are retained but not forced into the practice lane.

Table 1. Curtis production state on 7 May 2026.

| Signal                    | Current value                                                                            |
|---------------------------|------------------------------------------------------------------------------------------|
| Primary source            | YouTube channel @nalalan                                                                 |
| Inventory                 | 207 indexed videos                                                                       |
| Practice candidates       | 102 videos                                                                               |
| Long-form candidates      | 82 videos                                                                                |
| Media samples             | 20 captured samples                                                                      |
| Reviewed sections         | 51 sections                                                                              |
| Curtis-focused findings   | 80 findings                                                                              |
| Latest dated practice log | 5-1-26                                                                                   |
| Review model              | GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification |
| Current piece             | Confirmed title requires source window and verified piece-identification evidence        |
| Curtis-level completion   | Withheld when the available evidence only identifies repertoire                          |
| Current focus             | Audition-style setup consistency                                                         |
| Current constraint        | Open strings only, mirror-check contact point                                            |
| Boundary                  | Curtis admission cannot be predicted from current samples                                |

2.2 Review evidence

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The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

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3 First readout

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The first live readout is experimental but usable. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples and stored section reviews. The home screen now reduces the current state to a repertoire-identification state, a score state, one immediate tip, and the highlight clip that produced the label. This source-linked layout is deliberate: a detected title is not treated as self-evident unless the

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adjacent clip supports it, and a readiness percentage is not shown when the evidence only identifies repertoire. A visible repertoire list stores piece-level and day-level progress, and a dated practice log shows which video window was sampled for each practice day. Generic etude/caprice evidence remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from the stored review findings, piece record, sample index, and current progress plan.

**Table 2.** Current evidence classes and limits.

| Class         | Stored signal                                                   | Limit                                                         |
|---------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| Audio         | Tone, intonation, shifts, time, articulation                    | Some audio reviews fail or cannot inspect the excerpt cleanly |
| Video         | Setup, posture, bow lane, visible hand/bow evidence             | Still frames can hide motion, sound, and transition accuracy  |
| Metadata      | Dates, duration, channel, view count, practice-candidate labels | Metadata cannot judge playing quality by itself               |
| Progress plan | One focus, practice constraint, short session plan              | The plan is a next experiment, not an outcome prediction      |

## 4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification and evidence quality. YouTube metadata can identify candidate videos, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. A better system would combine reliable excerpt extraction, audio fingerprinting, score/title context, visual posture analysis, and longitudinal trend scoring.

The broader experiment is whether a practice system can make musical development legible over time.

A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

## 5 Materials and methods

### Backend

The backend is a FastAPI service deployed on Railway. It exposes `https://curtis.aolabs.io/api/curtis/media-status` for compact state, `https://curtis.aolabs.io/api/curtis/ops-check` for fuller operational state, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

### Source scan

The default public source is `https://www.youtube.com/@nalalan`. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The scan does not itself inspect performance quality.

### Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, early setup windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes,

and high-evidence forensic candidates can be seeded as comparison targets rather than accepted directly. Possible labels are treated as unidentified. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. Piece records retain day-level progress entries and source-window fields so the home screen can show the score state, a main highlight clip, and the longer-running piece record together.

## Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

## 6 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The useful claim is therefore limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

## References

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